

## Homework 1: Convex Sets and Functions

**Instructions:** Submit a single Jupyter notebook (.ipynb) of your work to Canvas by 11:59pm on the due date. All math must be formatted using LaTeX. **Be sure to show all the work involved in deriving your answers! If you just give a final answer without explanation, you may not receive credit for that question.**

You may discuss the concepts with your classmates, look on the internet for help, or ask an LLM for advice, but first try hard to solve problems on your own, and always write up the answers entirely on your own. Make sure you understand the concepts and solutions—don't just look up the answers or use AI to answer it for you. **Cite any sources you used outside of the class material (webpages, LLMs, etc.).**

1. **Dual Norms** Let  $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$  be a norm. Define the dual norm, denoted  $\|\cdot\|_*$ , as follows for any vector  $z \in \mathbb{R}^n$ :

$$\|z\|_* = \sup\{z^T x : \|x\| \leq 1\}.$$

- (a) Show that  $\|\cdot\|_*$  is a norm, i.e., show it satisfies the four requirements to be a norm that we covered in class.

**Hint:** You may use the following fact (without proving it yourself). Let  $S$  be a set and  $f : S \rightarrow \mathbb{R}, g : S \rightarrow \mathbb{R}$ . Then,

$$\sup\{f(s) + g(s) : s \in S\} \leq \sup\{f(s) : s \in S\} + \sup\{g(s) : s \in S\}$$

- (b) Show that the dual of the dual of a norm is the original norm. That is, for any norm  $\|\cdot\|$ , we have

$$\|x\|_{**} = \|x\|, \quad \text{for all } x \in \mathbb{R}^n.$$

- (c) Show that the dual of the  $L^2$  norm (a.k.a. Euclidean norm) is again the  $L^2$  norm. That is,

$$\|x\|_{2*} = \|x\|_2, \quad \text{for all } x \in \mathbb{R}^n.$$

**Hint:** You might find the Cauchy-Schwarz inequality useful!

**Not required—don't turn in!** If you want to go further, you might be interested to prove to yourself the following facts (or look up the proofs):

- The dual of the  $L^1$  norm is the  $L^\infty$  norm.
- The dual of the  $L^p$  norm is the  $L^q$  norm, where  $\frac{1}{p} + \frac{1}{q} = 1$ . This follows from a relation known as Holder's Inequality, which is a generalization of the Cauchy-Schwarz Inequality. See Appendix A.1.6 (dual norms) and Section 3.1.9 (Holder's Inequality) of the B&V book and also the Wikipedia pages on "Dual Norm," " $L^p$  Space," and "Holder's Inequality" for more information.

### 2. Point on a Hyperplane Closest to the Origin

Let  $a \in \mathbb{R}^n, a \neq 0$ , and  $b \in \mathbb{R}$ . Define the hyperplane

$$S = \{x \in \mathbb{R}^n : a^T x = b\}.$$

Give an expression for the point  $x_0 \in S$  that is closest to the origin in Euclidean distance, or said another way, the point where the  $L^2$  norm is minimal:

$$x_0 = \arg \min_{x \in S} \|x\|_2.$$

**Hint:** The formula for  $x_0$  should be an expression in terms of  $a$  and  $b$ .

### 3. Intersection of Hyperplanes

Consider two hyperplanes, defined by  $a_1, a_2 \in \mathbb{R}^n, a_1 \neq 0, a_2 \neq 0$ , and  $b_1, b_2 \in \mathbb{R}$ :

$$S_1 = \{x \in \mathbb{R}^n : a_1^T x = b_1\},$$

$$S_2 = \{x \in \mathbb{R}^n : a_2^T x = b_2\}.$$

- (a) Under what conditions are these hyperplanes disjoint, i.e.,  $S_1 \cap S_2 = \emptyset$ ? (State the conditions precisely in terms of the  $a_i$  and  $b_i$ .)
- (b) When  $S_1$  and  $S_2$  do intersect, what kind of space is  $S_1 \cap S_2$ ? What is its dimension? Please list all possible cases (**hint:** there are two cases, with two different dimensions), and describe the conditions under which each case happens.

### 4. Convex Cones from Convex Sets

Let  $S \subset \mathbb{R}^n$  be a convex set. Define the set

$$C = \{\theta x : x \in S, \theta \geq 0\}.$$

Show that  $C$  is a convex cone.

### 5. Convex Combinations and Convex Functions

- (a) Let  $C \subset \mathbb{R}^n$  be a convex set, and let  $f : C \rightarrow \mathbb{R}$  be a convex function. Show that for any  $x_1, \dots, x_k \in C$  and any  $\theta_1, \dots, \theta_k \geq 0$ , with  $\theta_1 + \dots + \theta_k = 1$ ,

$$f(\theta_1 x_1 + \dots + \theta_k x_k) \leq \theta_1 f(x_1) + \dots + \theta_k f(x_k).$$

- (b) Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be a convex function. Show that for any  $x_1, \dots, x_k \in \mathbb{R}^n$  and any  $\theta_1 > 0$ , and  $\theta_2, \dots, \theta_k \leq 0$ , with  $\theta_1 + \dots + \theta_k = 1$ ,

$$f(\theta_1 x_1 + \dots + \theta_k x_k) \geq \theta_1 f(x_1) + \dots + \theta_k f(x_k).$$

(**Careful:** Pay close attention to the direction of the various inequality signs!)

### 6. Convex Function or Not?

For each of the following functions,  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ , state whether it is convex, concave, both, or neither. Justify your answer in each case.

$$(a) f(x) = x^T x - \frac{1}{n}(1^T x)^2$$

$$(b) f(x) = \|x\|_1 + \frac{1}{2}\|x\|_2^2$$

$$(c) f(x) = \log \frac{\exp(a^T x + b)}{1 + \exp(a^T x + b)}, \quad \text{for } a \in \mathbb{R}^n \setminus \{0\}, b \in \mathbb{R}$$

### 7. Epigraphs of Functions

Problem 3.6 from B&V: When is the epigraph of a function a halfspace? When is the epigraph of a function a convex cone? When is the epigraph of a function a polyhedron?